IF-Matching: Towards Accurate Map-Matching with Information Fusion

Gang Hu, Jie Shao, Fenglin Liu, Yuan Wang, and Heng Tao Shen

Abstract—With the advance of various location-acquisition technologies, a myriad of GPS trajectories can be collected every day. However, the raw coordinate data captured by sensors often cannot reflect real positions due to many physical constraints and some rules of law. How to accurately match GPS trajectories to roads on a digital map is an important issue. The problem of map-matching is fundamental for many applications. Unfortunately, many existing methods still cannot meet stringent performance requirements in engineering. In particular, low/unstable sampling rate and noisy/lost data are usually big challenges. Information fusion of different data sources is becoming increasingly promising nowadays. As in practice, some other measurements such as speed and moving direction are collected together with the spatial locations acquired, we can make use of not only location coordinates but all data collected. In this paper, we propose a novel model using the related meta-information to describe a moving object, and present an algorithm called IF-Matching for map-matching. It can handle many ambiguous cases which cannot be correctly matched by existing methods. We run our algorithm with taxi trajectory data on a city-wide road network. Compared with two state-of-the-art algorithms of ST-Matching and the winner of GIS Cup 2012, our approach achieves more accurate results.

Index Terms—GPS trajectories, road network, map-matching, information fusion

1 Introduction

In recent years, with the maturity of various location-acquisition technologies there is an increasing popularity of recording the positions of various moving objects. Map-matching deals with the problem of matching a GPS trajectory to roads on a digital map. It becomes an important research issue, as it is a fundamental pre-processing step for many services, such as path planning [1], real-time navigation [2] and popular route finding [3]. However, the raw coordinate data collected from GPS sensing devices cannot always reflect the real positions of moving objects. There are a number of factors leading to such uncertainty. Lu et al. [4] listed the uncertainty in trajectory based work and proposed a visualization method to analysis uncertainty in trajectories. The data errors normally can be classified into two categories, namely, *measurement error* and *sampling error* [5].

The measurement error can be reasonably described as a Gaussian distribution. If we regard the space as an X-Y plane and the sampling position is at the origin, the distribution of real location is an ellipse or a circle on the X-Y plane, which is up to the variance and expectation of each dimensionality and the correlation coefficient of the two dimensionalities. Fig. 1 illustrates the measurement error. The height represents the possibility of the real position. That is, we can use a function of Gaussian distribution to represent the location possibility.

The sampling error is related to the sampling rates. It reflects the uncertainty of an object's moving track. In order to demonstrate the effect of sampling rate on the imprecision of the interpolated trajectory data, let us consider two sampling positions p_1 and p_2 of a moving object. If the sampling time interval is t and the average speed of the moving object is v, the moving (Euclidean) distance between the two positions is $l = v * t$. The possible position of the moving object between the two sampling positions is within an ellipse. As illustrated in Fig. 2, the foci of the ellipse are p_1 and p_2, the eccentricity $2c$ is the Euclidean distance between p_1 and p_2, and length of the major axis $2a$ is the maximum moving (Euclidean) distance l. Obviously, the real position is related to the form of the ellipse which is up to the sampling rate and the moving speed. Traditional approaches only find a moving track approximately according to the road topology, and they cannot deal with the sampling error. Only a few methods infer the moving track with the help of global relationship such as [3], [6]. However, their results still cannot meet stringent performance requirements of many applications, especially in difficult and complex environments such as dense urban areas. In our work, we employ the moving speed and moving direction as auxiliary information to infer real moving tracks. The sampling error then can be solved easily.

A number of methods have been proposed for map-matching, especially for the case of low sampling rates such as [7], [8], [9], and noise such as [8]. However, although these methods may perform well on some particular datasets, they need strict conditions (such as particular range of sampling rates and regularity of road network). The real condition could be very complex. We take the taxi trajectory

- *G. Hu, J. Shao, and F. Liu are with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China. E-mail: {hugang, lfl}@std.uestc.edu.cn, shaojie@uestc.edu.cn.*
- *Y. Wang is with the Department of Industrial and Systems Engineering, National University of Singapore 117576, Singapore. E-mail: iseway@nus.edu.sg.*
- *H. T. Shen is with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China, and the School of Information Technology and Electrical Engineering, University of Queensland, Brisbane 4072, Australia. E-mail: shenht@itee.uq.edu.au.*

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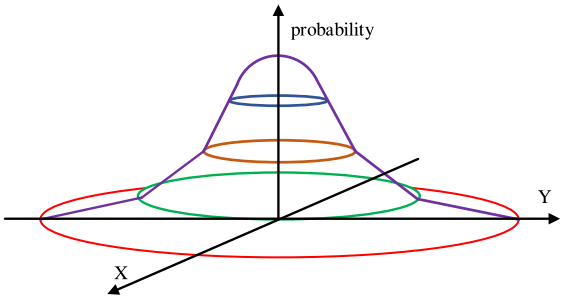

Fig. 1. An illustration of measurement error.

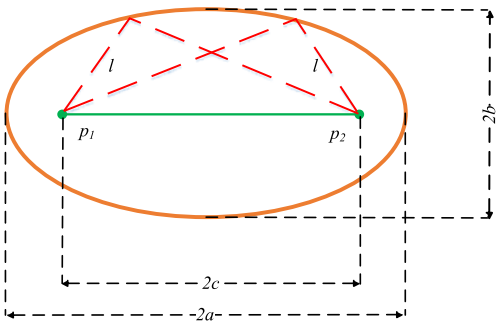

Fig. 2. An illustration of sampling error.

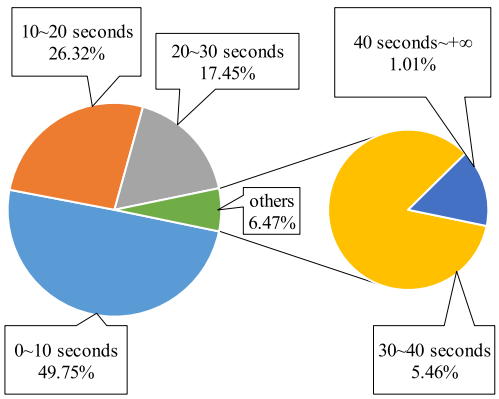

Fig. 3. Sampling rate distribution of the taxi trajectories collected in Chengdu.

dataset collected in Chengdu, China for an analysis in the following.

First, for saving the energy cost and communication, the taxis usually report their locations to the dispatching center with *low and unstable sampling rates*. The sampling rates usually vary from a few seconds to minutes. Yuan et al. [10] presented the sampling rates of taxi GPS positions in Beijing, and we made statistics of sampling rates for our dataset. The sampling rates distribute as Fig. 3. Similar observations can be made that the sampling rates vary greatly and most of the data belong to the low frequency.

Second, due to a variety of factors such as sensor failure or transmission error, the sampling data reported to the dispatching center cannot be fully used due to *partial data loss*. If we consider the simplest case where the sampling data tuple contains three fields of *longitude*, *latitude* and *timestamp*, a full tuple should include all the three fields but in practice longitude, latitude or timestamp may be missing. The full tuple ratio decreases with the increase of the number of data fields. As for the taxi data of Chengdu (about 2.21 billion tuples in

TABLE 1
Percentage of Data Loss for Different Fields

missing	location	timestamp	direction	speed	any
ratio	32.77%	12.47%	15.77%	29.81%	46.73%

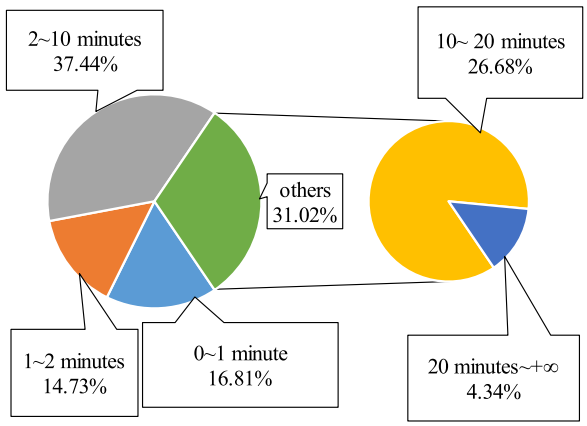

Fig. 4. After dropping incomplete data, sampling rate distribution of the taxi trajectories collected in Chengdu.

TABLE 2
Partial Fields of Sensed Taxi Data in Chengdu

longitude	latitude	timestamp	direction	speed	state

total), we made statistics about the full tuple with *location*, *speed*, *direction* and *timestamp*. Table 1 shows the ratio of data loss. Without recovering the data but dropping those incomplete tuples, the available data will be more sparse. As Fig. 4 shows, 37.44 percent of the sampling rates belong to the range of 2-10 minutes. The sampling rates falling in 10-20 minutes are 26.28 percent and exceeding 20 minutes are about 4.34 percent. Thus, the taxi dataset has a low sampling rate and it is a challenge to map-matching.

In order to monitor real-time state and provide more useful information, a taxi usually not only reports the GPS position but also speed, direction and so on. Table 2 presents partial sensed taxi data of Chengdu. All the information can be useful for map-matching. Traditional map-matching which only considers the location coordinates may not work well in complex situations such as intensive road, low sampling rate and data loss. We take the characteristic of real data into account to build a holistic model using a tuple $< longitude, latitude, timestamp, speed, direction >$ for describing a moving object, and propose an algorithm for map-matching with the model. The moving direction can be of great importance to select a road. We give an example in Fig. 5. Considering the moving direction and the road direction, clearly road 2 should be selected. The moving speed is also an important factor for decision, especially when traffic jams occur. Our approach better handles different complex situations in reality such as intensive road, low sampling rate and data loss, and thus can achieve a more accurate map-matching.

Overall, our approach is based on the following insights:

- *Information Fusion*. We use multiple information to describe a moving object and adopt the model for

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Fig. 5. Usage of direction for matching.

map-matching. Fusing multiple information can play an important role in the selection of a road, particularly in ambiguous cases.

- *Global Effect*. A point has relationship with the position context, and a moving object can be affected by other objects moving with it. The former is reflected by that we take the global relationship into consideration. Which road a point shall be mapped to will depend on the whole point trajectory. The latter is reflected by the consideration of moving speed. The objects moving together shall have similar moving speeds. Thus, it is a key for road selection if a traffic jams occurs.

Based on these observations, we propose an algorithm called *IF-Matching* with emphasis on information fusion. It has the following advantages:

- Our approach not only considers the spatial geometric and topological structures of road network, but also the status of moving objects. Thus, it will be more practical and reasonable.
- Our approach contains both spatial and temporal analysis, and thus can adapt to the reality of complex situations.
- Experimental study shows that IF-Matching achieves more accurate results compared with two state-of-the-art map-matching algorithms, ST-Matching [9] and the winner of GIS Cup 2012 map-matching task [11].

The rest of paper is organized as follows. We formulate the problem and give an overview of our solution in Section 2. Detailed discussion and analysis of IF-Matching algorithm are presented in Sections 3 and 4. The results of experimental study are reported in Section 5. We review related work in Section 6 and finally, we conclude in Section 7.

2 Preliminaries

In this section, we define some basic concepts, and give an overview of our IF-Matching algorithm.

2.1 Problem Formulation

Definition 1 (Sampling Tuple). *A sampling tuple S represented by $< location, direction, speed, timestamp >$ is a sampling point of a moving object that contains location (which usually consists of latitude and longitude), moving direction, moving speed and timestamp.*

For representing direction, a base direction and a positive direction are required. We define the north as the base direction and the clockwise as the positive direction. The direction is the angle between the base direction and the moving direction. As Fig. 6 shows, Y-axis denotes the base direction and the moving direction of a point S is $\overrightarrow{d}$. As we defined above, the moving direction can be represented as the angle θ.

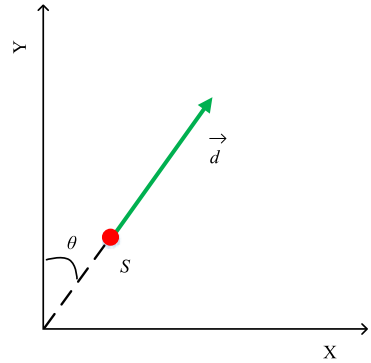

Fig. 6. Representation of direction.

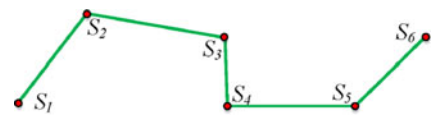

Fig. 7. Sampling trajectory.

Definition 2 (Sampling Trajectory). *A sampling trajectory T is a sequence of sampling tuples with a time interval between two consecutive sampling tuples not exceeding a certain threshold ΔT, i.e., $T : S_1 \longrightarrow S_2 \longrightarrow \cdots S_i \cdots \longrightarrow S_n$ where $0 < S_{i+1}.t - S_i.t < \Delta T$ and $1 \leq i < n$. An example of sampling trajectory is shown as Fig. 7.*

For the taxi data in this work, the distribution of sampling rate is shown as Fig. 4. Thus, we have to focus on the low sampling rate trajectories with $\Delta T \geq 5$ minutes.

Definition 3 (Road Segment). *A road segment r is a directed edge on road network with two terminal points $(r.s, r.e)$. Each road segment has a length $r.length$ and a speed constraint $r.speed$ which is the maximum speed allowed on this road segment.*

Definition 4 (Road Network). *A road network described as $G(V, E)$ is a directed graph, where V is a set of nodes representing intersections or terminal points, and E is a set of edges linking those nodes.*

Definition 5 (Route). *A route RO is a set of connected road segments on road network G, i.e., $RO : r_1 \longrightarrow r_2 \longrightarrow \cdots \longrightarrow r_k \cdots \longrightarrow r_n)$, where $r_{k+1}.s = r_k.e (1 \leq k < n)$. Obviously, the start point and end point of a route can be represented as $RO.s = r_1.s$ and $RO.e = r_n.e$.*

Definition 6 (Road Segment Projection). *The road segment projection of a sampling tuple S to a road segment r_i is a candidate point C_i (introduced in Definition 7) on r_i such that $C_i = \arg\min_{\forall C_i \in r_i} dist(r_i, S)$, where $dist(r_i, S)$ returns the distance between the sampling point and the road segment.*

Definition 7 (Candidate Points). *The candidate points C_i of a sampling tuple is a set of points that a sampling tuple projected to a set of road segments within a threshold R. The candidate point is illustrated as Fig. 8.*

Finally, the problem of map-matching is defined as follow:

Definition 8 (Map-Matching). *Given a sampling trajectory T and a road network $G(V, E)$, find the real route RO from G. The map-matching is illustrated as Fig. 9.*

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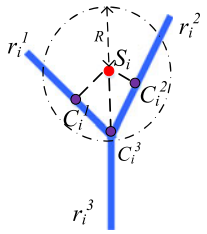

Fig. 8. Road segment projection and candidate points.

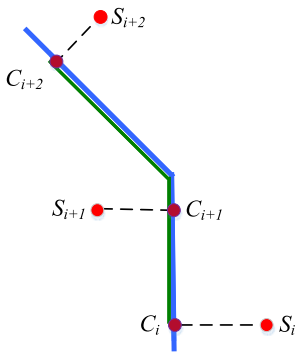

Fig. 9. An illustration of map-matching.

2.2 Our Model with Information Fusion

Previous work on map-matching regarded a moving object as a series of locations, which are only represented by latitude and longitude. In our opinion, the location information can only describe the static attributes ignoring the moving state. Actually, with the development of sensor technologies, more and more information can be recorded expediently, and fusion of multiple information sources becomes very promising. We propose a model to describe a moving object which includes four fields of *location*, *speed*, *direction* and *timestamp*. The model is shown as Fig. 10. Location is a moving object's current position, which usually is its GPS coordinates including the latitude and longitude. Speed is the moving object's current moving speed, which usually can be collected by the velocity transducer. Direction is moving object's current moving direction, which can be collected with the digital compass. Timestamp is the time of sampling reported to the management center.

Overall, the location captures the static properties and the speed and the direction capture the dynamic properties. In this way, a moving object will be represented more comprehensively. We will use this model for map-matching in our algorithm.

2.3 Algorithm Framework

Our IF-Matching algorithm is based on the above model which makes full use of information fusion of multiple data sources. Moreover, we consider all sampling points on a trajectory and all trajectories having relationship with the current trajectory. At the same time, we take the taxi driver's intelligence into consideration for matching to a humanistic route. In summary, our algorithm contains five phases: candidate preparation, history speed mining, surrounding speed estimation, spatial-temporal analysis and result matching. The framework of our algorithm is given as Fig. 11.

Candidate Preparation: The candidate preparation component is to execute the road segment projection. This component contains the road network $G < V, E >$, and will use

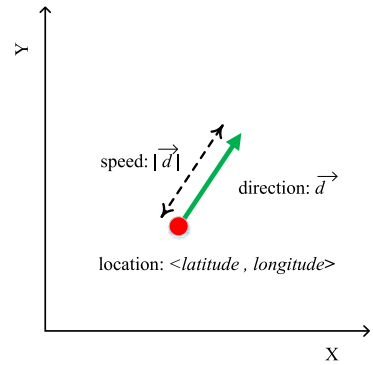

Fig. 10. Our proposed model.

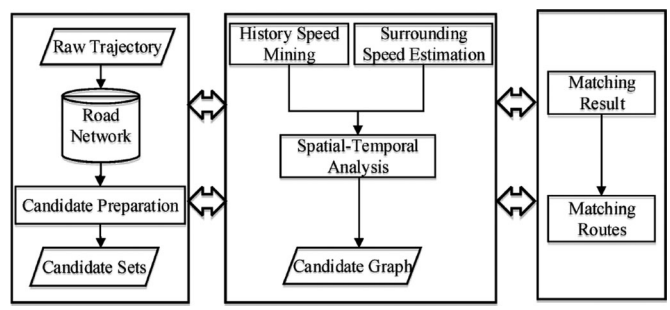

Fig. 11. Algorithm framework.

its topological information. The input of this component is a set of GPS sampling trajectories. The component will project the sampling tuples to G to get some candidates according to its projection rule. The output of this component is a set of candidates for each sampling tuple.

History Speed Mining: The history speed mining component is to get a historical, estimated speed which is a function of time. The history speed will be used by the spatial-temporal analysis. In our opinion, the speed pattern can be described by a model and the model can be used as a reference to match a raw trajectory to its real route. In this manner, the historical information will be a factor to decide the matched route.

Surrounding Speed Estimation: The surrounding speed estimation component is to compute the current surrounding speed. The real position of a sampling point is not only related to the points that the sampling point located on trajectory, but also has relationship with its surroundings. For example, if a taxi runs with a low speed, it may be located on a crowded road or low speed highway, but not a smooth highway. This component will help to filter some roads. It will be useful for selection of candidate road segments. Similar with history speed mining component, this component is an independent unit that we only use a method to achieve its function and do not care how to achieve it efficiently. It can be regarded as an offline component.

Spatial-Temporal Analysis: The spatial-temporal analysis is a component to match a sampling trajectory T to a road network $G < V, E >$ for getting a route RO. The spatial analysis is followed by the temporal analysis.

- Spatial analysis considers the distance between an individual sampling point and its candidate road segments, the topological information of the road network, moving objects' intelligence, and the moving

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direction. To avoid roundabout routes, shortest path is used to measure the similarity between each candidate route and the "true" route, because we assume the moving object will always follow the "shortest path" to move. However, such an assumption may result in filtering out some correct "strange candidate points" which do not meet the requirements of the shortest path. Fortunately, on account of the moving direction, the final route is not always to select the shortest path, but the correct point can be reserved.

- Temporal analysis considers the actual moving speed. Comparing the moving speed with the history speed and the surrounding speed, we can distinguish spatially similar routes which cannot be processed by spatial analysis. As a result, a more realistic candidate point will be selected.

The output of this component is a candidate graph. The nodes of this graph include a set of candidate points for each sampling point, and the edges of this graph include a set of routes (may not be the shortest path) between any two neighboring candidate points. The nodes and edges are all assigned weight values based on the results of spatial-temporal analysis.

Result Matching: This component evaluates the candidate graph using the weight information assigned during spatial-temporal analysis. It matches the given trajectory to a route with highest score in the candidate graph.

3 IF-Matching Algorithm

In this section, we describe our algorithm in detail. An example is also given in the end to demonstrate how IF-Matching algorithm works.

3.1 Candidate Preparation

Given a sampling trajectory $T = S_1 \longrightarrow S_2 \longrightarrow \cdots S_k \cdots \longrightarrow S_n$, we first perform the procedure of road segment projection by retrieving a set of candidate road segments within radius R for each sampling tuple S_i $(1 \leq i \leq n)$. Then, we compute candidate points at each candidate road segment. The method to select candidate points is as follows: if the projection of the sampling tuple onto the road segment is between its endpoints, choose the geometry projection as the candidate point; otherwise, choose the endpoint which is closer to the sampling tuple with regard to the Euclidean distance.

As Fig. 8 shows, given a candidate selection radius R, the sampling tuple S_i will have three candidate road segments r_i^1, r_i^2 and r_i^3. After selecting candidate points, S_i will select C_i^1, C_i^2 and C_i^3 as its candidates. Algorithm 1 shows the detailed procedure. Fig. 12 gives an example of a sampling trajectory projection. After the projection of sampling tuples, we obtain candidate road segments (CRS) and candidate points (CP) for each S_i. $CRS_1 = \{r_1^1, r_1^2\}$ and $CP_1 = \{C_1^1, C_1^2\}$. The candidate trajectory is represented as a sequence of candidate points $CP_1 \longrightarrow CP_2 \longrightarrow CP_3$.

3.2 History Speed Mining

In history speed mining, we use the historical data to build a model for speed. This component is based on the assumption that moving objects usually follow regular speed changes. The historical information can be a reference speed.

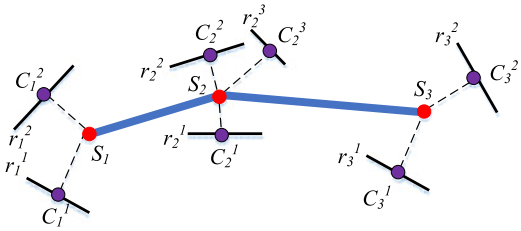

Fig. 12. Candidate road segments and candidate points.

Algorithm 1. Candidate Projection

Input:
road network $G < V, E >$, sampling tuple S.
Output:
list of projected candidate points C or projection failure.
Description:
$dist(r_i, S) = getDistance(r_i, S)$; //get distance of a sampling tuple to a road segment
if $dist(r_i, S) > R$ **then**
  report projection failure; //drop r_i
else
  if $isBeyondRoad$ **then**
    $C.Add(r_i.e)$; //select road segment endpoints
  else
    $C.Add(C_i)$; //select the projection points
  end if
end if
return C;

In real-life, each road has a speed constraint. However, the speed constraint can only limit a vehicle to move below a speed and usually is useless for us. For example, an urban road's maximum speed is 60 km/h, and a vehicle on it usually runs at 40 km/h without traffic. Unfortunately, the vehicles can only drive at 10-20 km/h during traffic peak. Fig. 13 shows the actual speed distribution for a road with speed constraint of 60 km/h in our dataset. Thus, a model for speed is needed.

We build the history speed model by time of a day. We divide one day into 12 parts by time, i.e., each part has two hours. In daily life, the traffic peak usually lasts about two hours. We compute the average speed for each part on each road segment. For each input of time, we do not plan to only use its interval's speed (because this will decrease the availability of data greatly and ignore the global influence). We also consider speeds of other time intervals to evaluate an interval's speed with a weighting strategy. Thus, the speed of an input time T is defined as

$$V_{(T)} = \frac{\sum_{i=1}^{12} v_i \times \left(1 - \frac{|T - t_i|}{24}\right)}{12}, \tag{1}$$

where t_i is the center of a two-hour interval (1,3,5, . . . ,21,23) and v_i is the corresponding speed of that interval. After this step, a history speed can be obtained as a reference for temporal analysis. Algorithm 2 shows the detailed procedure.

3.3 Surrounding Speed Estimation

In surrounding speed estimation, we take the road condition into consideration for speed. This component is based on that moving speed is affected by actual road condition. The

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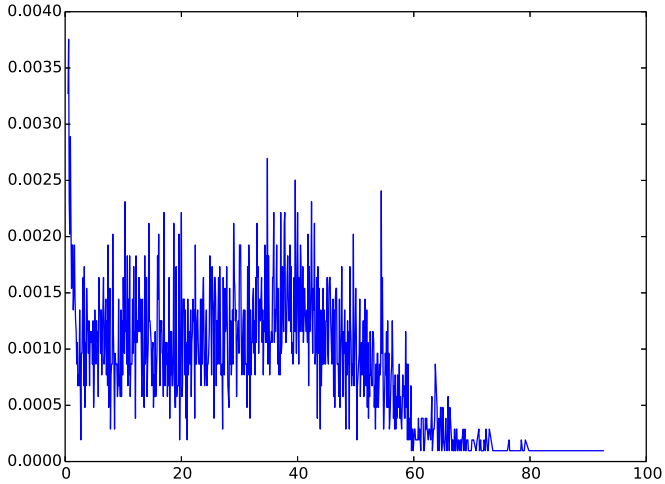

Fig. 13. Actual speed distribution of vehicles for a road with speed constraint of 60 km/h in our dataset. *X*-axis represents speed and *Y*-axis represents proportion of vehicles.

influence can be reflected by other moving objects on the same road. That is to say, the moving objects which locate on a same road can be treated as an whole and have a similar speed trend. Their speeds tent to converge to a central value.

Algorithm 2. History Speed Mining

Input:
road network $G < V, E >$, a set of projected tuples SS.
Output:
a road graph $G' < V, E' >$ with history speed.
Description:
1: **for all** road segment r_i in G **do**
2: select the projected tuples SS_i in SS for each r_i;
3: compute the history speed by time for each road segment;
4: add the history speed to G;
5: **end for**
6: **return** G';

We describe our idea as follow. First, we find candidate road segments for each sampling tuple by the rule of candidate preparation described in Section 3.1. At the same time, the candidate points can be found. Second, we find the moving objects located on the same road segment within a time threshold Δt. Moreover, these moving objects have a similar moving direction. Because usually a road is two-way, the opposite moving direction has no relationship with the sampling tuple. As Fig. 14 shows, the starting point S_s and the ending point S_e are two terminal points of a route, S_s is located in a residential area, and S_e is located in a commercial area. Traffic jam will occur at 8:00 am, because many people will go to work from S_s to S_e. The vehicles from S_s to S_e will run very slowly, but the vehicles drive following opposite direction from S_e to S_s will run relatively fast, because only a few people will drive from S_e to S_s. However, this situation will turn into the opposition at 6:00 pm, because people will go home. Third, we compute the center speed for each candidate road segment. The surrounding speed is defined as

$$V_{(v)} = \frac{\sum_i^n v_i}{n}, \tag{2}$$

Fig. 14. Residential area and commercial area.

where v_i is the moving speed selected by step two and n is the number of selected objects. After this step, a surrounding speed is obtained as a reference for temporal analysis. The detailed procedure is showed as Algorithm 3.

Algorithm 3. Surrounding Speed Estimation

Input:
candidate road segments r, a set of sampling tuples SS_T.
Output:
a surrounding speed $V(v)$ for each candidate road segment.
Description:
1: **for all** candidate road segments r_i in r **do**
2: select the projected tuples SS_{Ti} in SS_T for each r_i;
3: compute the surrounding speed for each road segment $V(v)$;
4: **end for**
5: **return** $V(v)$;

3.4 Spatial Analysis

In spatial analysis, we use geometric and topological information of the road network, moving objects' intelligence and the moving direction to evaluate candidate points found in the previous component. The geometric information is incorporated by using *observation probability*, the topological information and the moving objects' intelligence are expressed by using *transmission probability*, and the moving direction is described by using *selection probability*.

Definition 9 (Observation Probability). *If a sampling tuple S_i matches a candidate point C_i^j, the observation probability is defined based on the computed distance between the two points $dist(C_i^j, S_i)$.*

As described in Section 1, the measurement error can be reasonably described as a two-dimensional normal distribution about the distance between C_i^j and S_i. In this work, we assume that the two perpendicular directions are uncorrelated and have a same distribution. Thus, the error measurement can be described as normal distribution $N(\mu, \delta^2)$. Formally, we define observation probability $N\left(x_i^j\right)$ of S_i as

$$N\left(x_i^j\right) = \frac{1}{\delta\sqrt{2\pi}} e^{-\frac{\left(x_i^j-\mu\right)^2}{2\delta^2}}, \tag{3}$$

where $x_i^j = dist(C_i^j, S_i)$ is the distance between C_i^j and S_i. Based on empirical evaluation, we use a zero-mean normal distribution with a standard deviation of 20 meters.

The observation probability only considers the measurement error, so it cannot be the final decision. We further introduce the transmission probability as follows:

Definition 10 (Transmission Probability). *For two candidate points C_{i-1}^j and C_i^k of two neighboring sampling tuples*

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S_{i-1} and S_i respectively, the transmission probability from C_{i-1}^j to C_i^k is defined by the probability that the real route from S_{i-1} to S_i follows the shortest path from S_{i-1} to S_i.

In this step, we take the moving objects' intelligence into consideration. We assume a moving object always follows the shortest path. Thus, we compare the distance of two sampling tuples with the shortest path between candidate points. The transmission probability is computed as

$$W\left(C_{i-1}^j \rightarrow C_i^k\right) = \frac{d_{(i-1)\rightarrow i}}{w_{(i-1,j)\rightarrow(i,k)}}, \tag{4}$$

where $d_{(i-1)\rightarrow i} = dist(S_i, S_{i-1})$ is the Euclidean distance between S_i and S_{i-1}, and $w_{(i-1,j)\longrightarrow(i,k)}$ is the length of the shortest path from C_{i-1}^j to C_i^k. In the next, we will add the direction as a factor to select candidate points. The selection probability is defined as follows:

Definition 11 (Selection Probability). *Given a sampling tuple and its candidate points, the selection probability will reflect the probability that each candidate point is selected as the "true" point.*

Comparing a moving object's moving direction to corresponding candidate road segment's direction, we can easily determine a road segment. Recall the example shown in Fig. 5. Obviously, road 2 is more likely to be the real road segment the sampling tuple located on than road 1, because the moving direction of S is more similar with road 2 than road 1. Thus, we compute selection probability as

$$D\left(C_i^j\right) = 1 - \frac{\arccos\frac{\vec{D_c}\cdot\vec{D_s}}{|\vec{D_c}||\vec{D_s}|}}{2\pi}, \tag{5}$$

where $\vec{D_S}$ is the moving object's direction vector, and $\vec{D_C}$ is the candidate road segment's direction vector.

Combining Equations (3), (4), and (5), we define the spatial analysis function $S(C_{i-1}^j \rightarrow C_i^k)$ as the product of observation probability, transmission probability and selection probability:

$$S\left(C_{i-1}^j \rightarrow C_i^k\right) = N\left(x_i^j\right) \cdot W\left(C_{i-1}^j \rightarrow C_i^k\right) \cdot D\left(C_i^j\right), \tag{6}$$

where C_{i-1}^j and C_i^k are any two candidate points for two neighboring sampling tuples S_{i-1} and S_i respectively. Equation (6) computes the *likelihood* that an object moves from C_{i-1}^j to C_i^k using the product of three probability functions. Thus, geometric and topological information, moving objects' intelligence and the direction are all taken into consideration.

With spatial analysis, a set of candidate routes from C_{i-1}^j to C_i^k are generated for any two neighboring sampling tuples S_{i-1} and S_i. Each route will have a spatial measurement value computed from Equation (6).

3.5 Temporal Analysis

In temporal analysis, we use the history speed and the surrounding speed as a reference to evaluate a candidate. This component will further assist spatial analysis to distinguish two or more similar roads.

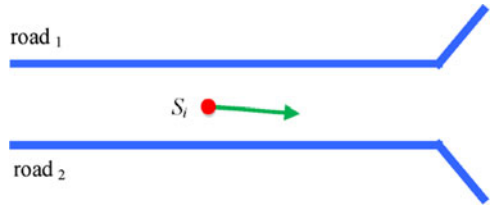

Fig. 15. Using temporal analysis for distinguishing similar roads.

As Fig. 15 shows, there are two similar roads, which have an equal speed constraint. Assume the physical condition of road 1 is better than road 2. Thus, objects on road 1 move faster than those on road 2 due to different road conditions. The spatial analysis cannot handle this situation alone. The speed constraint cannot distinguish this situation as well. Fortunately, when considering the history speed and surrounding speed, road 1 will be selected as the more suitable matching result.

The key idea of temporal analysis is to take the speed into consideration. As a result of history speed mining and surrounding speed estimation, a reference speed is obtained. What we should do is to compare the moving object's speed with the reference speed. The reference speed is to be calculated with history speed and surrounding speed as follow:

$$V_{ref} = \alpha \cdot V_{(T)} + \beta \cdot V_{(v)}, \tag{7}$$

α and β are two parameters which are the weighting factors. $V_{(T)}$ is the history speed, $V_{(v)}$ is the surrounding speed, and V_{ref} is the result.

The history speed is always available from historical data. However, surrounding speed depends on the availability of other traffic samples. It is possible that the surrounding speed information is unavailable due to data sparsity. To make the reference speed always available, there is a constraint as follow:

$$\alpha + \beta = 1. \tag{8}$$

Now, we compare the moving object's speed with the reference speed to find the road with similar speed. The temporal analysis function is defined as follow:

$$T\left(C_{i-1}^j \rightarrow C_i^k\right) = \frac{\sum_{i=1}^n \left(r.v_{ref}^i \times \bar{v}_{(i-1,j)\rightarrow(i,k)}\right)}{\sqrt{\sum_{i=1}^n \left(r.v_{ref}^i\right)^2} \times \sqrt{\sum_{i=1}^n \left(\bar{v}_{(i-1,j)\rightarrow(i,k)}\right)^2}} \tag{9}$$

$r.v_{ref}^i$ represents the reference speed of road segment r and $\bar{v}_{(i-1,j)\rightarrow(i,k)}$ represents the average moving speed and contains k elements of the same value. $T(C_{i-1}^j \rightarrow C_i^k)$ represents the result of temporal analysis function.

With temporal analysis, a temporal measurement value is computed from Equation (9). In the next, the temporal measurement value will be used as auxiliary information to find a route.

3.6 Result Matching

With the above processes, a candidate point set can be obtained for each sampling tuple and two measurements of spatial analysis and temporal analysis have been calculated for each candidate road segment. In this section, we

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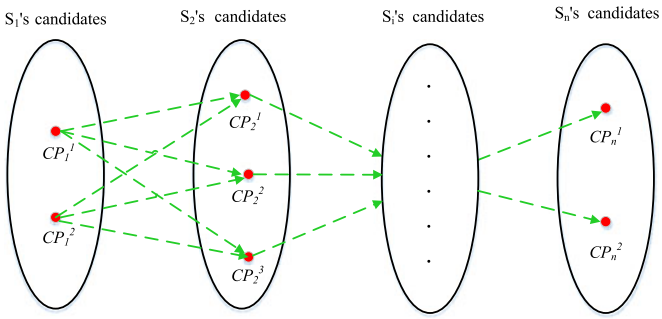

Fig. 16. Candidate graph.

describe how to confirm a particular point among the candidate points and to find a matched route for a trajectory.

After candidate preparation, spatial analysis and temporal analysis, a new candidate graph $G_c(V_c, E_c)$ has been built for a trajectory $S_1 \longrightarrow S_2 \longrightarrow \cdots \longrightarrow S_n$. V_c is the set of candidate points and E_c is the set of edges with weight values that need to be computed. As Fig. 16 shows, C_i^j represents the candidate points for sampling tuple S_i, the edge C_{i-1}^j to C_i^k represents a candidate route with two weights $S(C_{i-1}^j \rightarrow C_i^k)$ and $T(C_{i-1}^j \rightarrow C_i^k)$. There are $n_1 \times n_2 \times \cdots \times n_n$ candidate routes (n_i is the number of candidates for each sampling tuple). Now what we should do is to find a best one as the final route. We define IF (information fusion) function for C_{i-1}^j to C_i^k as

$$IF\left(C_{i-1}^j \rightarrow C_i^k\right) = S\left(C_{i-1}^j \rightarrow C_i^k\right) \times T\left(C_{i-1}^j \rightarrow C_i^k\right). \quad (10)$$

In Equation (10), $S(C_{i-1}^j \rightarrow C_i^k)$ represents the result of spatial analysis, $T(C_{i-1}^j \rightarrow C_i^k)$ represents the result of temporal analysis. The edge C_{i-1}^j to C_i^k represents a candidate route with a weight $IF(C_{i-1}^j \rightarrow C_i^k)$. Now, the graph $G' < V', E' >$ has been build completely. We need to find a particular route for a sampling trajectory. A score function $IF(T_c)$ for a candidate trajectory $T_c : C_1^j \longrightarrow C_2^k \longrightarrow \cdots \longrightarrow C_m^n$ is defined as follow:

$$IF(T_c) = \sum_{i=2}^{m} IF\left(C_{i-1}^j \rightarrow C_i^k\right). \quad (11)$$

$IF(C_{i-1}^j \rightarrow C_i^k)$ is the IF function, m is the number of sampling tuples and j, k are the numbers of candidate points. In IF-Matching, the trajectory with the highest overall score is selected as the best matched route. More formally, the final matched route RO for a sampling trajectory T is selected as

$$RO = \arg\max IF(T_c). \quad (12)$$

The algorithm of IF-Matching is given as Algorithm 4. First, we compute the candidate points and road segments for each sampling tuple by Algorithm 1 and a history speed mining by Algorithm 2. Second, we conduct spatial and temporal analysis. Finally, we find a result for each sampling trajectory.

To demonstrate the working of our algorithm, we give an example shown as Fig. 17. First, we get the candidates for each sampling tuple. In this example, the candidates of S_1 are $\{CP_1^1, CP_1^2\}$, the candidates of S_2 are $\{CP_2^1, CP_2^2, CP_2^3\}$ and the candidates of S_3 are $\{CP_3^1, CP_3^2\}$. Next, we calculate IF scores.[1] This process is showed as Tables 3, 4, and 5. The IF score $IF(CP_1^1 \longrightarrow CP_2^1 \longrightarrow CP_3^1)$ = 0.70 + 0.01 + 0.04 =0.75. The IF score $IF(CP_1^1 \longrightarrow CP_2^1 \longrightarrow CP_3^2)$ = 0.70 + 0.01 + 0.02 =0.73. The IF score $IF(CP_1^1 \longrightarrow CP_2^2 \longrightarrow CP_3^1)$ = 0.70 + 0.06 + 0.07 =0.83. Finally, we get the route with the highest IF score as 0.83. By a backtracking step (line 27 of Algorithm 4), we can get the route is $CP_1^1 \longrightarrow CP_2^2 \longrightarrow CP_3^1$.

Algorithm 4. IF-Matching

Input:
road network $G < V, E >$, a set of projected tuples S_m, a sampling trajectory $T : S_1 \rightarrow S_2 \rightarrow \cdots \rightarrow S_n$;
Output:
a matched route $ROList : ro_1 \rightarrow ro_2 \rightarrow \cdots \rightarrow ro_n$;
Description:
1: **for all** sampling tuple S_i in T **do**
2: $CP_i = CandidatesPreparation(G, T)$;
3: $G' = HistorySpeedMining(G, S_i)$;
4: **end for**
5: **for all** CP_1^j in CP_1 **do**
6: $F[] = N(CP_1^j)$; //$F[]$ denotes the highest score computed so far
7: **end for**
8: **for** $i = 2$ to n **do**
9: **for all** CP_i^j in CP_i **do**
10: max $= -1$;
11: **for all** CP_{i-1}^k in CP_{i-1} **do**
12: $mid = F[CP_{i-1}^k] + IF[CP_{i-1}^k \longrightarrow CP_i^j]$;
13: **if** (mid > max) **then**
14: max $= mid$;
15: $PRE[CP_i^j] = max$; //$PRE[]$ denotes the parent of current candidate point
16: **end if**
17: $F[CP_i^j] =$ max;
18: **end for**
19: **end for**
20: **end for**
21: $IF = \arg\max(F[CP_n^j])$;
22: **for** $i = n$ to 2 **do**
23: $ROList.add(CP_i^m)$; //$ROList$ donates a list of matched points in reverse order
24: $CP_{i-1}^x = PRE[CP_i^m]$;
25: **end for**
26: $ROList.add(CP_1^m)$;
27: $ROList.reverse()$;
28: **return** $ROList$;

4 Algorithm Analysis

In this section, we analyze the time complexity of IF-Matching algorithm. The time complexity of our algorithm is divided into three parts, including *candidate graph construction*, *spatial analysis* and *temporal analysis*. In order to analyze the time complexity, we use the following notations to denote some variants. n denotes the number of sampling tuples on a given trajectory, and m denotes the number of road segments in the road network. Meanwhile, let k denote the maximum number of candidates for a sampling tuple.

1. Actually all 12 candidate routes need to be computed. However, for ease of description only 3 routes are shown, including the one with the highest score.

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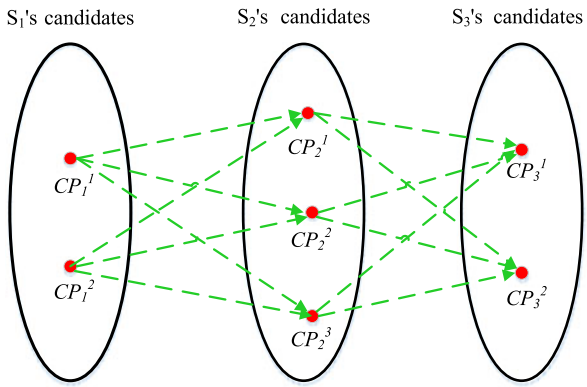

Fig. 17. Result matching example.

TABLE 3
Observation Probability N of CP_1

CP_1^1	CP_1^2
0.70	0.60

First, there are n sampling tuples. The maximum number of candidates for a sampling tuple is k and each candidate can reach its neighboring candidates. Thus, the maximum number of shortest paths is $(n-1)k^2$ in our algorithm. During the candidate graph construction process, the time complexity is $O(n*k*2m*logm)$. During the spatial analysis, the time complexity is $2(n-1)k^2+(n-1)$ and the temporal analysis part is $(n-1)k^2$. Thus, we can find that the total time complexity is $O(n*k^2*m*logm+n*k^2)$ on the worst case. However, the complexity is only $O(n*m*logm)$ in the best case.

In our approach, a large number of shortest path calculations are needed. Unfortunately, it is a costly process for getting results. There are many methods to get those shortest paths such as Dijkstra, A*, SPFA and Bellman-Ford. However, all of them are of high computation expense and will affect our algorithm's efficiency. In our implementation, we compute some shortest paths in advance for some important routes and save to an index table. Thus, when we need the shortest path, we can look up at the index table. If the result can be found, we can apply it to our algorithm. Otherwise, we will compute the shortest path and save it to the index table. As for the history speed, we can compute it in advance. The surrounding speed is also dealt with the same manner, and updated with a fixed time interval. In this way, our algorithm will focus on the matching process.

5 Experiment

In this section, we report our experimental evaluation. We will verify that our method is practical and can achieve better accuracy than previously proposed methods. First, we describe the setting of our experiment. We evaluate the performance of map-matching with a real dataset of taxi trajectories in Chengdu, China. Thus, we will present the dataset comprehensively. Then, two evaluation criteria are introduced. We will compare the proposed IF-Matching with the ST-Matching algorithm [9] which is a simply but efficient method, and another algorithm proposed by Tang et al. [11] which wins the champion of GIS Cup 2012 [12].

TABLE 4
IF Score of $CP_1^i \longrightarrow CP_2^j$

	CP_2^1	CP_2^2	CP_2^3
CP_1^1	0.01	0.06	0.04
CP_1^2	0.02	0.03	0.09

TABLE 5
IF Score of $CP_2^k \longrightarrow CP_3^l$

	CP_3^1	CP_3^2
CP_2^1	0.04	0.02
CP_2^2	0.07	0.03
CP_2^3	0.01	0.08

5.1 Experimental Setting

5.1.1 Dataset

Our dataset is obtained from a taxi company which collects the taxi data in Chengdu, China all the time. Each taxi has a number of sensors to capture the corresponding information. However, the data collected from sensors often have inevitable errors due to physical constraints. Although there are some methods which may be applied to correct the data errors in the first place, we do not employ this approach but adopt the original data directly. We want to demonstrate that IF-Matching is still able to perform well on a raw dataset.

Road Network: In our experiment, we use the map which covers Chengdu, China as the underlying road network. The map is represented with the shape file which can be opened with professional GIS tools such as QGIS. We visualize the Chengdu road network in Fig. 18 by the tool of QGIS Desktop 2.10.1 with GRASS 6.4.3 for Windows. The road network is formed by linking some road segments (edges). It contains 24,702 nodes and 33,532 edges. Each road segment is stored in the form exemplified in Table 6.

As Fig. 18 shows, the road network of Chengdu is not as regular as some other cities such as Beijing. The irregularity could bring additional challenges. In map-matching, a road intersection is a key point to link many roads. Therefore, intersection points will largely affect the quality of map-matching. We analysis the degree of each node in the road network of Chengdu. As Fig. 19 shows, road intersections occupy a large proportion.

The Chengdu road network is not regular and there are many tortuous paths and side ways. In order to display the characteristic, we measure the length of each road segment. The result is shown in Fig. 20. We can see that, the road segment lengths vary from 0 to 500 m. Almost half of them belong to 400-500 m, and many of them belong to 0-100 m. The above two figures can help us understand the irregularity of the road network.

Taxi Data: The taxi dataset contains 13,993 taxis' trajectories in Chengdu for one month. The total number of attributes for each data record is 15. The sampling data record can be exemplified in Table 7. Only a part of the attributes are useful for map-matching, and the other attributes can be ignored. We also obtained nine taxis' trajectories which recorded the real position of each sampling point as groundtruth.

The nine trajectories cover both downtown region and suburban region, with a total length of up to 388 kilometers

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Fig. 18. Road network of Chengdu.

over more than 14 hours. The number of original sampling points is up to 2864, and Table 8 shows the number of sampling points for each ground truth trajectory after dropping incomplete sampling tuples. Each trajectory has a different number of sampling points, and the total number of sampling points is 435.

There are lots of incomplete sampling tuples in this dataset. In order to display the characteristic of noisy/lost data, we made statistics about the taxi data loss shown in Table 1. As Fig. 4 shows, after dropping those incomplete tuples, the average sampling rate is actually quite low. Fig. 21 is the taxi speed distribution in the Chengdu dataset. The taxi speeds are mostly below 60 km/h and only a few belong to the range of 60 to 100 km/h. Specially, the speed belonging to 0-20 km/h is up to 49.72 percent. This suggests that the speed constraint is not very meaningful for map-matching. That is why we need to compute the history speed and the surrounding speed in our approach. As mentioned above, Fig. 3 gives the sampling rate distribution of the original data. We can see that the sampling time intervals are almost between 0 to 10 s, which differers from the dataset of Beijing. To a certain extent, the original dataset has many high sampling data. However, after dropping those incomplete tuples, the distribution of sampling interval is shown as Fig. 4 and many of them belong to low sampling data.

In our method, we need to select candidates for each sampling point. Thus, we made statistics about number of candidates for each sampling point. Fig. 22 is the distribution of candidate numbers for all test trajectories, with the radius R selected as 100 m. Most of those sampling points have more than 20 candidates. In our method, the number of candidates will determine the accuracy and efficiency.

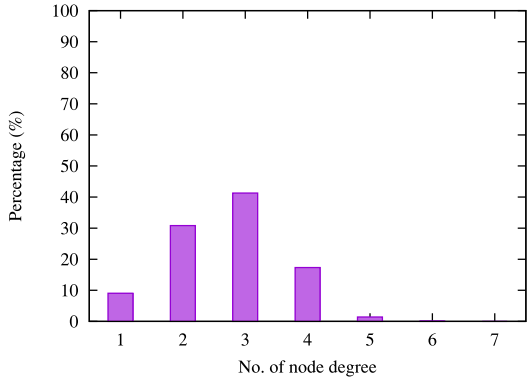

Fig. 19. Distribution of node degrees.

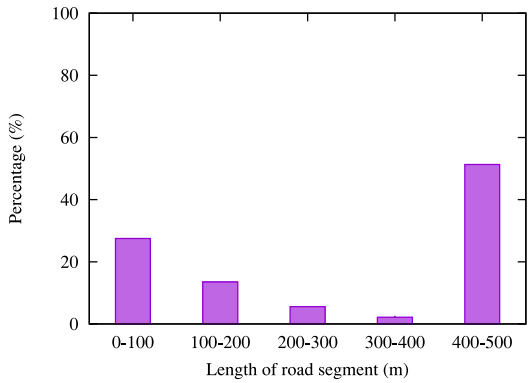

Fig. 20. Distribution of road segment lengths.

Too many candidates will affect efficiency, and too few candidates will affect accuracy.

5.1.2 Parameters

In our work, a few parameters need to be set, including the radius of candidates R (in candidate preparation), the expectation μ and standard deviation δ of Gaussian distribution (in spatial analysis), and α and β for reference speed (in temporal analysis). We will tune these parameters according to the performance study based on our accuracy evaluation defined next.

5.1.3 Evaluation Criteria

The accuracy of map-matching is measured by three criteria A_r, A_l and A_p, calculated by following equations:

$$A_r = \frac{number\ of\ correctly\ matched\ road\ segments}{number\ of\ all\ road\ segments\ of\ a\ test\ trajectory}.$$

A_r focuses on the number of correctly matched road segments. A larger A_r means better performance.

TABLE 6
The Form of Road Segment Data

ROAD	*ROAD_ID*	*NAME_CHN*	*NAME_PY*	*FNODE*	*TNODE*	*LENGTH*
2,523	13	1/2	*XiaJie*	36,210,836	36,210,846	111.800003
...	...	...	...	...	...	...
WIDTH	*ROAD_CLASS*	*S_LINES*	*P_LINES*	*N_LINES*	*MAX_SPEED*	*DIRECTION*
6	45,000	2	1	1	40	1
...	...	...	...	...	...	...

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TABLE 7
The Form of Taxi Data

$VP_TERMINALID$	$VP_VEHICLENUM$	$VP_LONGITUDE$	$VP_LATITUDE$	$VP_DIRECTION$
100,200,102,246	$**A-TC012$	103.996794	30.628516	15.000000
...	...	...	...	...
VP_SPEED	$VP_LOCATIONTIME$	$VP_DEVICESTATUS$	$VP_ALERTFLAG$	$VP_DRIVETI$
0.0	2014/08/01	01 : 11 : 42	131	0
...	...	...	...	...
$VMEALERT$	$VP_TOTALDRIVETIMEALERT$	$VP_INCOMEFLAG$	$VP_NORTHFLAG$	$VP_EASTFLAG$
0	0	1	0	0
...	...	...	...	...

$$A_l = \frac{\sum length\ of\ correctly\ matched\ road\ segments}{length\ of\ a\ test\ trajectory}.$$

A_l is the ratio of matched route length and real trajectory length. A larger A_l means better performance.

$$A_q = \frac{distance\ between\ real\ position\ and\ matched\ position}{length\ of\ the\ road\ segment\ containing\ real\ position}.$$

A_q focuses on the difference between the real position and matched position. A smaller A_q means better performance.

While A_r and A_l reflect the overall matching accuracy, A_q is sensitive to *matched points*, and it reflects the algorithm performance for individual points. If A_r and A_l are small, A_q will make no sense. Therefore, we only use A_r and A_l to measure the performance with varying parameters. Finally, we will use A_r, A_l and A_q to measure the performance of our approach and compared methods with optimal parameters.

5.2 Experimental Results

5.2.1 Tuning δ in Observation Probability

First, we study the effect of the parameters in observation probability. In our work, we assign the candidate point as the center of Gaussian distribution (i.e., μ is 0), and δ is standard deviation. In this experiment, we fix other parameter values by assigning the candidate selection radius R as 100 meters, α as 0.4 and β as 0.6.

As Fig. 23 shows, when δ is smaller than 20 meters A_r and A_l increase with δ increasing. A_r and A_l decrease with δ increasing when δ is larger than 20 meters. When δ is small, the probability that nearby candidate points are selected as matched points will increase, which leads to matching error. When δ is large, the probability that nearby candidate points are selected as matched points will decrease, and the effect of observation probability will weaken. This experiment confirms that the zero-mean normal distribution with a standard deviation of 20 meters is appropriate for our algorithm.

5.2.2 Tuning Candidates Selection Radius R

Candidate selection radius R determines the candidate points and candidate road segments. Small R may result in few candidate points and road segments, while large R corresponds to many candidates which is expensive for computation.

As Fig. 24 shows, A_r and A_l increase with R increasing and become stable when R exceeds 100 meters. When R is small, the road segments covering real positions could be missing. With R exceeds 100 meters, almost all the road segments covering real positions will be selected as candidate road segments. Thus, we set 100 meters as the candidate selection radius in our algorithm.

5.2.3 Turning α and β in Reference Speed

In temporal analysis, a reference speed is needed to select candidates. The reference speed is composed by history speed and surrounding speed. As Equations (7) and (8) show, large α emphasizes the influence of history speed and large β emphasizes the influence of surrounding speed. We

TABLE 8
Number of Sampling Points for Each Groundtruth Trajectory

trajectory	1	2	3	4	5	6	7	8	9
number	10	14	19	28	39	67	78	84	96

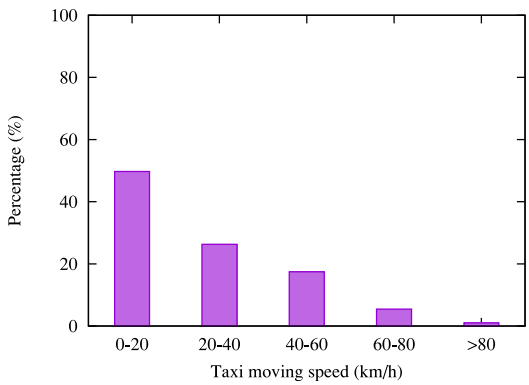

Fig. 21. Distribution of taxi speeds.

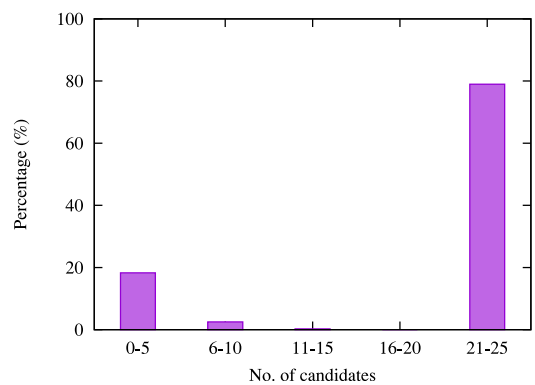

Fig. 22. Distribution of candidate numbers.

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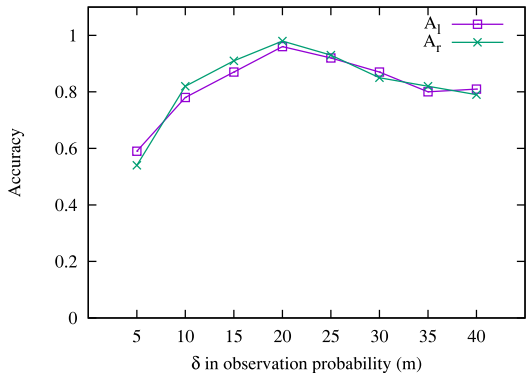

Fig. 23. Effect of δ in observation probability.

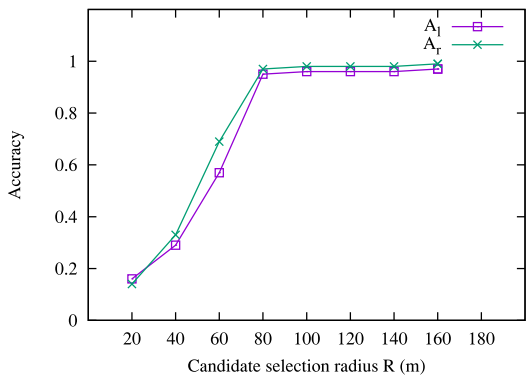

Fig. 24. Effect of candidate selection radius R.

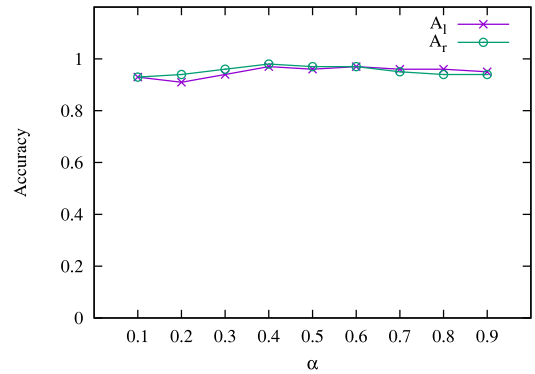

Fig. 25. Effect of α in reference speed.

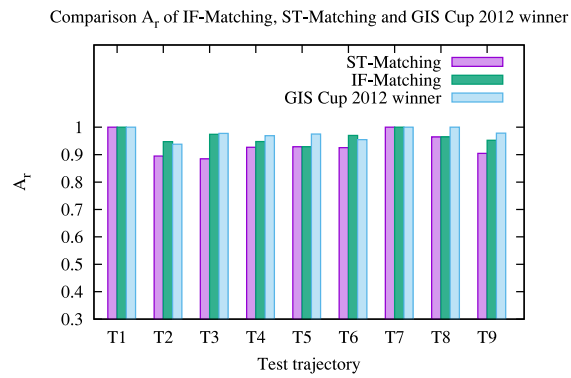

Fig. 26. The result of A_r.

study the effect of α and β on A_r and A_l shown as Fig. 25. Varying α seams to only have a slight influence on matching result. This can be explained as the reference speed is only used to distinguish similar roads (as shown in Fig. 15) or faulty candidates which cannot be identified in spatial analysis. The highest values of A_r and A_l are obtained when α is 0.4. Therefore, α is assigned as 0.4 by default.

5.2.4 Matching Result Comparison

Fig. 26 is the result of A_r with each of the nine trajectories to test different algorithms. From Fig. 26, we can find our IF-Matching algorithm gets a better accuracy compared with the ST-Matching algorithm in terms of A_r. It exceeds 90 percent, even reaches 100 percent. It is worthwhile to mention that, although it seems that the algorithm of GIS Cup 2012 winner is perfect compared with IF-Matching, actually the reason is that A_r only considers the correctly matched road segments. The algorithm of GIS cup 2012 winner matches all sampling points to endpoints for its strategy of candidate selection. In this way, their algorithm only selects the endpoints as candidate points to index a road segment. However, we preferentially select projected points as candidate points. If there is no projected points for a sampling point, endpoints will be selected as candidate points. The strategy to select candidate points and candidate road segments is presented in Section 2.

The reason why we select candidates in such a way is that, we want to match a sampling tuple to road network as accurately as possible. For an endpoint belonging to two or more road segments, it cannot reflect real position on a road segment. Although consecutive endpoints can infer a trajectory, *exact* location on a road is of great of importance for many applications. In that sense, IF-Matching actually performs as well as GIS Cup 2012 winner in terms of A_r. Meanwhile, there are two more accuracy criteria, A_l and A_q which we defined above and need to be considered.

A_l is the ratio of matched route length and real trajectory length. Compared with A_r, it also reflects the overall matching accuracy, but is evaluated by length rather than quantity. As Fig. 27 shows, we can find that our algorithm outperforms ST-Matching and the GIS Cup 2012 winner in terms of A_l. In other words, the matched routes for a sampling trajectory have some differences for three matching algorithms and the result of our approach is more precise.

A_q is sensitive to difference between the real position and matched position. It can better reflect the matched result for each individual sampling point. As Table 9 shows, our approach outperforms ST-Matching and GIS Cup winner in terms of A_q. The result confirms the matched result of our approach is closest to real position for individual sampling points. ST-Matching takes a similar approach with our algorithm to select candidates. Therefore, the ST-Matching algorithm achieves better A_q than GIS Cup 2012 winner.

In summary, the proposed IF-Matching performs better not only in terms of a overall matched route, but also can match real position more accurately for individual sampling points.

6 Related Work

Computing with moving object trajectories have attracted lots of research attention recently, such as using clustering [13] for analysis [14], and hot spot discovery [15] and recommendation [16]. If road network is available, normally it is

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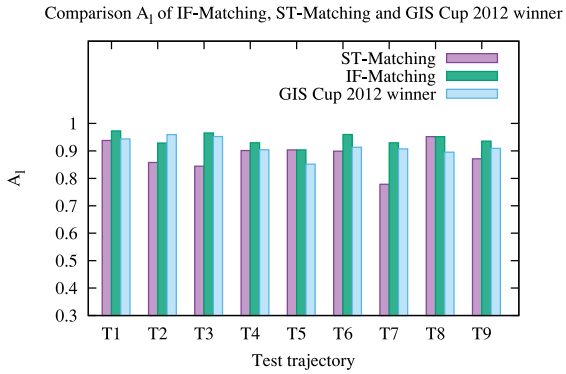

Fig. 27. The result of A_l.

better to integrate positioning data with roadway centrelines for identifying the correct link on which a moving object is travelling. A number of map-matching algorithms have been developed using different techniques [17]. Many conventional map-matching methods employ *local* or *incremental* algorithms to map current or neighboring positions onto road segments on a map. The incremental matching such as [18] pursues the local matching of a small portion of the trajectory. This approach is fast in computation and performs well when sampling frequency is very high. However, its performance is sensitive to the decrease of sampling frequency, and sometimes results in a suboptimal solution. On the other hand, *global* algorithms such as [19] batch process the entire input trajectory with the road network. IF-Matching takes the global approach and achieves better accuracy at a higher computational cost.

Map-matching algorithms can also be classified into four groups: geometric, topological, probabilistic, and other advanced techniques. A geometric map-matching algorithm makes use of the geometric information of the road network by considering only the shape of the links [18], [20]. It does not consider the way links are connected to each other. A topological map-matching algorithm further makes use of the connectivity and contiguity of the links [21]. An enhanced probabilistic map-matching algorithm considers various error sources associated with the navigation sensor and the road network data quality [22]. A weighting based and trajectory interpolation based map-matching algorithm is proposed in [23] to calibrate large-scale vehicle trajectories. Other advanced algorithms use Kalman filter [24], fuzzy logic [25], or the application of Hidden Markov Model [8]. Variants of HMM based algorithms such as [9], [10] have been adopted for their abilities to concurrently evaluate multiple hypotheses of the actual mapping in order to find the eventual maximum likelihood solution. Our proposed method is inspired by the model of HMM. However, these researches have not employed heading and speed information *directly* as in our work. Each measured point consists of only a time-stamped latitude/longitude pair, such as the test data used in ACM SIGSPATIAL GIS Cup 2012 [12].

A few studies in the literature utilizing the additional information (e.g., speed and direction) for map-matching have been noticed, and they can be divided into two categories. The first is to infer such additional information by location data. In [26], [27], [28], speed information (inferred from the distance between consecutive points and the time interval) is used to compare with the constraint speed of road segment to determine a candidate road segment. In [29], moving direction information (inferred from sampling point sequence) is used to compare with the curvedness of road to determine a candidate road segment, and build a model for estimating the traversal time to distinguish the shortest or fastest path. The second is to leverage additional information directly collected by sensors. In [30], the sensed moving speed and direction are used only to find a set of road segments for sampling points. Different from our work, it does not consider how to generate the weight of each candidate road segment and estimate the matched result. In addition, global effect of all trajectories and surrounding sampling points has not considered.

TABLE 9
The Result of A_q

approach	IF-Matching	ST-Matching	GIS Cup 2012 winner
A_q	0.11	0.19	0.31

Besides the spatial geometric and topological structures of road network, we utilize the status of moving objects. With information fusion, IF-Matching achieves better results on geometrically noisy and temporally sparse trajectories.

7 Conclusion

In this paper, a holistic model is introduced to describe a moving object by incorporating more sensed information. Based on this model, we design a framework utilizing information fusion for map-matching. In the proposed IF-Matching algorithm, we take the advantages of moving speed and direction, and global effect. We conduct experiment with a real dataset and compare our approach with two competitive algorithms ST-Matching and GIS Cup 2012 winner. The result shows that IF-Matching performs better than these methods.

Acknowledgments

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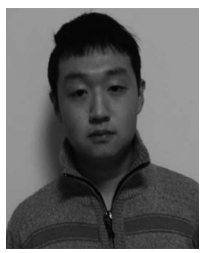
Gang Hu is currently working toward the master's degree in the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research interests include spatial query processing and trajectory mining.

Jie Shao received the BE degree in computer science from Southeast University, Nanjing, China, in 2004 and the PhD degree in computer science from the University of Queensland, Brisbane, Australia, in 2009. He is currently a professor in the Big Media Computing Center, School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. He worked as a research fellow with the University of Melbourne, from 2008 to 2011, and with National University of Singapore, from 2012 to 2014. His research interests include spatial databases and multimedia information retrieval.



Fenglin Liu is currently working toward the undergraduate degree in the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research interests include spatial query processing and multimedia databases.

Yuan Wang received the PhD degree in operations research, with a specialization in maritime transportation optimization and simulation, from the National University of Singapore. She is a research fellow in the Department of Industrial and Systems Engineering, National University of Singapore. Her research interests include mathematical modeling, complex system simulation, and optimization heuristics. She is currently working on data driven and on-demand scheduling problems in the Center of Next Generation Logistics in National University of Singapore.

Heng Tao Shen received the BSc (Hons.) and PhD degrees from the Department of Computer Science, National University of Singapore, in 2000 and 2004, respectively. He joined the University of Queensland as a lecturer, senior lecturer, and reader, where he became a professor, in 2011. He is currently a professor with the Big Media Computing Center, School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research interests mainly include multimedia/mobile/web search, and big data management on spatial, temporal, multimedia, and social media databases. He is an associate editor of the *IEEE Transactions on Knowledge and Data Engineering*, and a PC co-chair of the ACM Multimedia 2015.

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